

SPhO 2011

1) To fall to 85% of its initial amplitude, one must have

$$\frac{A}{A_0} = 0.85 = e^{-\gamma t} \therefore t = -\frac{\ln 0.85}{\gamma} = -\frac{4(\ln 0.85)R^2\varrho}{9\eta}$$

We evaluate it:

$$t = -\frac{4(\ln 0.85)(5.2 \times 10^{-3})^2(0.89 \times 10^3)}{9(1.78 \times 10^{-5})} = \mathbf{98 \text{ s}}$$

And its error

$$\Delta t = \left| \frac{\partial t}{\partial \eta} \right| \Delta \eta + \left| \frac{\partial t}{\partial R} \right| \Delta R + \left| \frac{\partial t}{\partial \varrho} \right| \Delta \varrho = t \left(\frac{\Delta \eta}{\eta} + 2 \frac{\Delta R}{R} + \frac{\Delta \varrho}{\varrho} \right)$$

Which we can evaluate as

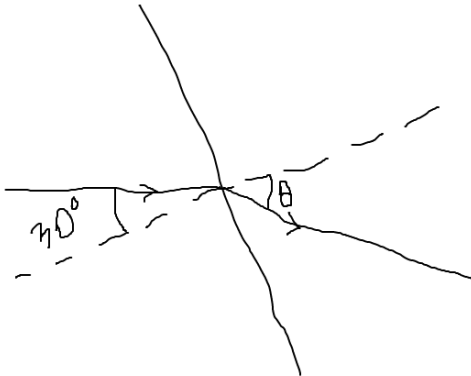
$$\Delta t = 98 \cdot \left(\frac{0.02}{1.78} + 2 \cdot \frac{0.2}{5.2} + \frac{0.05}{0.89} \right) = \mathbf{14 \text{ s}}$$

Of which the contributions by each parameter is

- $\left| \frac{\partial t}{\partial \eta} \right| \Delta \eta = t \frac{\Delta \eta}{\eta} = 1.1 \text{ s}$
- $\left| \frac{\partial t}{\partial R} \right| \Delta R = 2t \frac{\Delta R}{R} = 7.5 \text{ s}$
- $\left| \frac{\partial t}{\partial \varrho} \right| \Delta \varrho = t \frac{\Delta \varrho}{\varrho} = 5.5 \text{ s}$

of which the **parameter R contributes the biggest error** to the final result.

2) (a) We draw a picture of the refraction angle



And obtain (by Snell's Law) $\sin \theta = n \sin 30^\circ = 1.50 \cdot \frac{1}{2} = 0.75$.

Photons: $E = pc, E = \frac{hc}{\lambda}, p = \frac{h}{\lambda}$. Thus $\frac{dp}{dt} = \frac{1}{c} \frac{dE}{dt} = \frac{\text{Power}}{c}$

We are interested in the momentum transfer from the light beam to the prism in the positive y-direction, which is at a rate

$$\begin{aligned} F &= \frac{\Delta p}{\Delta t} = 2 \frac{P}{c} \sin(\theta - 30^\circ) = \frac{2P}{c} (\sin \theta \cos 30^\circ - \cos \theta \sin 30^\circ) \\ &= \frac{2(100)}{2.997 \times 10^8} \left(0.75 \cdot \frac{\sqrt{3}}{2} - \sqrt{1 - 0.75^2} \cdot \frac{1}{2} \right) = \mathbf{2.13 \times 10^{-7} \text{ N}} \end{aligned}$$

Which is precisely the upward force delivered by the light beam to the prism, and is thus the change (decrease) in weight of the prism.

(b) Each light beam exerts a force on the transparent sphere in the similar method as part a (transfer of momentum by refraction). This results in a potential gradient (attracting the sphere toward the light beam). At least two beams are needed so as to manipulate the potential field to move the sphere to the desired location (similar to (sticky) chopsticks).

(c) Because air is mostly nitrogen gas N_2 , we assume that air is made out of diatomic molecules ($\gamma = 7/5$) with estimated molecular mass $\mu = 28$ g/mol.

The energy of a monoatomic particle is $\frac{3}{2}kT$, a diatomic particle is $\frac{5}{2}kT$.

Note: let f be the number of degrees of freedom, then adiabatic constant $\gamma = 1 + \frac{2}{f}$, energy constant $\frac{f}{2}$. Don't mix these up.

$$\frac{3}{2}kT = \frac{1}{2}mv^2 \rightarrow v = \sqrt{\frac{3kT}{m}}$$

$$v_{sound} = \sqrt{\frac{5kT}{m_{N_2}}}$$

And we assume that the small smoke particles are in Brownian motion. They thus travel with speed

$$v_{smoke} = \sqrt{\frac{3kT}{m_{smoke}}}$$

$$\text{Thus } m_{smoke} = m_{N_2} \cdot \frac{3}{5} \left(\frac{v_{sound}}{v_{smoke}} \right)^2$$

$$\therefore m_{smoke} = 28 \cdot \frac{3}{5} \frac{(330)^2}{(0.10 \times 10^{-3})^2} = 1.82952 \times 10^{14} \text{ g/mol} = \mathbf{3.039 \times 10^{-10} \text{ g}}$$

3) (i) Conservation of energy $Fd = \frac{1}{2}mv^2$ so immediately before collision,

$$v = \sqrt{\frac{2Fd}{m}}$$

and because it is equal-mass elastic collision, the velocity immediately after collision is $v = 0$.

(ii) There is no force while the second bead travels so immediately before collision with third bead, velocity of second bead is

$$v = \sqrt{\frac{2Fd}{m}}$$

and because it is equal-mass elastic collision, the velocity immediately after collision with third bead is $v = 0$.

(iii) Constant acceleration $a = F/m$ this time interval between successive collisions is $t = \sqrt{2d/a} = \sqrt{2md/F}$. Average speed is

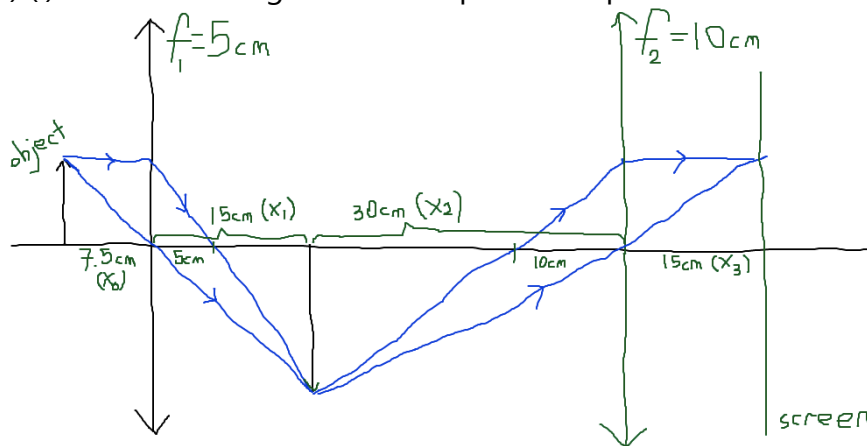
$$v_{ave} = \frac{1}{2} \sqrt{\frac{2Fd}{m}} = \sqrt{\frac{Fd}{2m}}$$

And the speed of the shock wave (follows the speed of one bead) is $\sqrt{\frac{2Fd}{m}}$.

(iii) The terminal speed of the shock wave will be such that the momentum transfer in the time interval between beads is equal to the amount of momentum needed to be transferred to the new attaching bead. (The first bead will eventually be arbitrarily heavy so the force will contribute to minimal acceleration in the long run).

$$p = Ft = F \cdot \frac{d}{v_T} = mv_T \therefore v_T = \sqrt{\frac{Fd}{m}}$$

4) (i) We draw a diagram of the optics set-up:



All lengths are in cm. The object is initially placed at distance $x_0 = 7.5$ to the left of the first lens. The (real) image is formed at a distance x_1 to the right of the first lens, and satisfies the lens equation

$$\frac{1}{x_0} + \frac{1}{x_1} = \frac{1}{f_1} \therefore x_1 = \frac{1}{1/f_1 - 1/x_0} = \frac{1}{1/5.00 - 1/7.50} = 15.00$$

This image is formed with magnification $M_1 = -x_1/x_0 = -15.00/7.50 = -2$ where the negative sign is due to the image being inverted.

In order to have final magnification on the screen equal to $+1.00$, we must have that the magnification of the first image through the second lens is $M_2 = \frac{+1.00}{M_1} = -0.5$. Thus if the first image is distance x_2 to the left of the second lens, and the final image on the screen is distance x_3 to the right of the second lens, then $x_3 = -M_2 x_2 = 0.5 x_2$ and the lens equation

$$\frac{1}{x_2} + \frac{1}{x_3} = \frac{3}{x_2} = \frac{1}{f_2} \therefore x_2 = 3f_2 = 30.00, \quad x_3 = 0.5x_2 = 15.00$$

The distance between the object and the screen is thus

$$d = x_0 + x_1 + x_2 + x_3 = 7.5 + 15.00 + 30.00 + 15.00 = \mathbf{67.50 \text{ cm}}$$

(ii) We need to solve the system of equations

$$\begin{cases} \frac{1}{x_0} + \frac{1}{x_1} = \frac{1}{f_1}, & f_1 = 5.00 \\ \frac{1}{x_2} + \frac{1}{x_3} = \frac{1}{f_2}, & f_2 = 10.00 \\ x_0 + x_1 + x_2 + x_3 = d, & d = 67.50 \\ (M_1 M_2 =) \frac{x_1 x_3}{x_0 x_2} = M, & M = +3.00 \end{cases}$$

the first two are from the lens equation, the third is by fixing of the image and screen positions, the last is by the request of the final magnification.

To solve the system of equations, we use the parameters x_0, x_2, M'_1, M'_2 where $M_1 = -M'_1, M_2 = -M'_2, x_1 = M'_1 x_0, x_3 = M'_2 x_2$. Then:

$$\begin{cases} x_0 = f_1 \left(1 + \frac{1}{M'_1} \right), & f_1 = 5.00 \\ x_2 = f_2 \left(1 + \frac{1}{M'_2} \right), & f_2 = 10.00 \\ x_0 + x_1 + x_2 + x_3 = d, & d = 67.50 \\ M'_1 M'_2 = M, & M = +3.00 \end{cases}$$

and the third equation becomes

$$f_1 \left(1 + \frac{1}{M'_1} \right) (1 + M'_1) + f_2 \left(1 + \frac{1}{M'_2} \right) (1 + M'_2) = d$$

Now we substitute in $M'_2 = M/M'_1$:

$$f_1 \left(1 + \frac{1}{M'_1} \right) (1 + M'_1) + f_2 \left(1 + \frac{M'_1}{M} \right) \left(1 + \frac{M}{M'_1} \right) = d$$

$$f_1 (M'_1 + 1)^2 + f_2 \left(1 + \frac{M'_1}{M} \right) (M'_1 + M) = d M'_1$$

$$\left(f_1 + \frac{f_2}{M} \right) M'^2_1 + (2f_1 + 2f_2 - d) M'_1 + (f_1 + M f_2) = 0$$

This is a quadratic in M'_1 . We substitute in the values of the coefficients:

$$\frac{25.00}{3} M'^2_1 - 37.50 M'_1 + 35.00 = 0$$

These give solutions

$$M'_{1a} = 3.1787, \quad M'_{1b} = 1.3213$$

and substituting into $M'_1 M'_2 = M$:

$$M'_{2a} = 0.9438, \quad M'_{2b} = 2.2705$$

substituting into the two equations $x_0 = f_1 \left(1 + \frac{1}{M'_1} \right)$ and $x_2 = f_2 \left(1 + \frac{1}{M'_2} \right)$:

$$\begin{aligned} x_{0a} &= 6.573, & x_{0b} &= 8.784 \\ x_{2a} &= 20.596, & x_{2b} &= 14.404 \end{aligned}$$

substituting into $x_1 = M'_1 x_0, x_3 = M'_2 x_2$:

$$\begin{aligned} x_{1a} &= 20.894, & x_{1b} &= 11.606 \\ x_{3a} &= 19.438, & x_{3b} &= 32.705 \end{aligned}$$

So there are evidently **two ways in which the lenses can be displaced**. For each way, we calculate the displacement of each lens from its initial position:

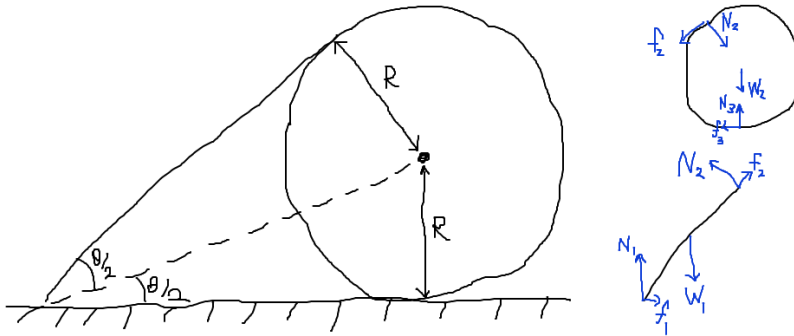
Way a:

- $\Delta x_{lens\ 1} = 7.50 - 6.573 = \mathbf{0.927\text{ cm to the left}}$
- $\Delta x_{lens\ 2} = 19.438 - 15.00 = \mathbf{4.438\text{ cm to the left}}$

Way b:

- $\Delta x_{lens\ 1} = 8.784 - 7.50 = \mathbf{1.284\text{ cm to the right}}$
- $\Delta x_{lens\ 2} = 15.00 - 14.404 = \mathbf{0.596\text{ cm to the right}}$

5) (i) We draw the free-body diagrams of the rod and the circle:



We have (due to static equilibrium) $F_{net,x} = 0$, $F_{net,y} = 0$ and $\tau = 0$ for both the rod and the stick. This gives us six equations to solve for the six unknowns $N_1, f_1, N_2, f_2, N_3, f_3$ (the weight $W_1 = \rho l g = \rho R g \cot \frac{\theta}{2}$ is known while W_2 is unknown to us, but fixed).

Taking torque on the rod about the ground contact point:

$$\frac{l}{2} W_1 \cos \theta = l N_2 \sin \theta \therefore N_2 = \frac{1}{2} \rho R g \cot \frac{\theta}{2} \cos \theta$$

Taking torque on circle about center:

$$f_2 R = f_3 R \therefore f_2 = f_3$$

Taking $F_{net,x} = 0$ for the circle:

$$N_2 \sin \theta = f_2 \cos \theta + f_3 = f_2 (1 + \cos \theta)$$

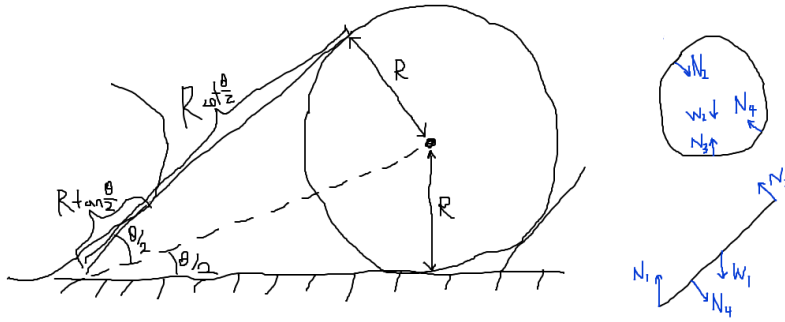
Thus

$$\frac{1}{2} \rho R g \cot \frac{\theta}{2} \cos \theta \sin \theta = f_2 (1 + \cos \theta)$$

we simplify the final expression:

$$f_2 = \frac{\rho R g \cot \frac{\theta}{2} \cos \theta \sin \theta}{2(1 + \cos \theta)} = \rho R g \frac{\cos \frac{\theta}{2} \cos \theta \left(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)}{\sin \frac{\theta}{2} \cdot 2 \left(2 \cos^2 \frac{\theta}{2} \right)} = \frac{\rho R g}{2} \cos \theta$$

(ii) We draw the diagram again, and revised free-body diagrams:



Here if there is any horizontal friction f_1 at the hinge similar to part i), it will cause $N_2 \neq N_4$, then the forces on the circle will not be balanced (where N_2 and N_4 are the only forces, collinear as the diameter), contradiction.

Applying $F_{net,x} = 0$ for the circle/rod:

$$N_2 \sin \theta = N_4 \sin \theta \therefore N_2 = N_4$$

Taking torque on the rod about the ground contact point:

$$\left(R \tan \frac{\theta}{2}\right) N_4 + \frac{R \cot \frac{\theta}{2}}{2} W_1 \cos \theta = \left(R \cot \frac{\theta}{2}\right) N_2$$

Thus

$$N_2 \left(1 - \tan^2 \frac{\theta}{2}\right) = \frac{1}{2} \rho R g \cot \frac{\theta}{2} \cos \theta$$

and simplifying the expression with $\tan \theta = 2 \tan \frac{\theta}{2} / (1 - \tan^2 \frac{\theta}{2})$:

$$N_2 = N_4 = \frac{\frac{1}{2} \rho R g \cot \frac{\theta}{2} \cos \theta}{1 - \tan^2 \frac{\theta}{2}} = \frac{1}{4} \rho R g \sin \theta \cot^2 \frac{\theta}{2}$$

6) Note: this question is missing data of the enthalpy of vaporisation of helium, $L_{He} = 20900 \text{ J/kg}$.

(i) The heat released by the lower half of the aluminium rod is

$$Q = mc_{al}\Delta T = \rho A \left(\frac{l}{2}\right) \cdot c_{al}(T_f - T_i) = 2700 \cdot (2.50 \times 10^{-4}) \cdot \frac{0.500}{2} \cdot 902 \cdot (300 - 4.20) = 45024 \text{ J}$$

this heat is transferred to vaporise the helium:

$$Q = m_{He}L_{He} = V_{He}\rho_{He}L_{He} \therefore V_{He} = \frac{Q}{\rho_{He}L_{He}} = \frac{45024}{125 \cdot 20900} = 0.0172 \text{ m}^3 = \mathbf{17.2 \text{ L}}$$

(ii) Heat conductivity k_{al} is related to heat flow rate via

$$\dot{Q} = k_{al}A \frac{dT}{dx}$$

Take note the conductivity given in the question is $31.0 \frac{\text{J}}{\text{s cm K}}$, so it is 3100 in SI units. Take note that only half the rod is under temperature gradient, the other half is completely at 4.2K, thus we only consider $\frac{l}{2}$:

$$\dot{Q} = k_{al}A \frac{\Delta T}{\Delta x} = (31.0 \times 10^2) \cdot (2.50 \times 10^{-4}) \cdot \frac{300 - 4.20}{0.500/2} = 916.98 \text{ J/s}$$

We also have

$$\dot{Q} = \dot{V}_{He}\rho_{He}L_{He} \therefore \dot{V}_{He} = \frac{\dot{Q}}{\rho_{He}L_{He}} = \frac{916.98}{125 \cdot 20900} = 0.000351 \text{ m}^3/\text{s} = \mathbf{0.351 \text{ L/s}}$$

7) When reflected by a medium of higher refractive index, the wave gains $+\pi$.

Thus $\frac{2\pi}{\lambda} \cdot (n \cdot 2t) = (2k + 1)\pi$. The first interference bands are at thickness $t = \frac{\lambda}{4n}$.

Note thickness is directly proportional to height from the top ($\frac{t}{h} = 2 \tan \frac{\theta}{2}$).

(i) The first red interference band is at

$$h_{red} = h_{violet} \cdot \frac{\lambda_{red}}{\lambda_{violet}} = 3.00 \cdot \frac{680}{420} = \frac{34}{7} = \mathbf{4.86 \text{ cm}}$$

(ii) The thicknesses are

$$t_{violet} = \frac{\lambda_{violet}}{4n} = \mathbf{78.75 \text{ nm}}$$

$$t_{red} = \frac{\lambda_{red}}{4n} = \mathbf{127.5 \text{ nm}}$$

(iii) The wedge angle is

$$\theta = 2 \arctan \frac{t}{2h} = \mathbf{1.504 \times 10^{-4} \text{ deg} = 2.625 \times 10^{-6} \text{ rad}}$$

8) The magnetic field due to a ring of radius R carrying current I at a distance x along the symmetry axis, is by Biot-Savart's Law:

$$dB = \frac{\mu_0}{4\pi} \frac{d\vec{l} \times \hat{r}}{r^2}$$

$$B = \left(\int_0^{2\pi} \frac{\mu_0}{4\pi} \cdot \frac{I r d\theta}{x^2 + R^2} \right) \left(\frac{R}{\sqrt{x^2 + R^2}} \right) = 2\pi \cdot \frac{\mu_0}{4\pi} \cdot \frac{I R x}{(x^2 + R^2)^{3/2}} = \frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}}$$

So we integrate this across the solenoid:

$$B = \int_{-l}^l \int_{R_1}^{R_2} \frac{\mu_0 n I R^2}{2(x^2 + R^2)^{3/2}} dR dx$$

Switch order of integrals and sub $x = R \tan \phi$, $dx = R \sec^2 \phi d\phi$, $\phi_0 = \arctan\left(\frac{l}{R}\right)$:

$$B = \int_{R_1}^{R_2} \int_{-\phi_0}^{\phi_0} \frac{\mu_0 n I R^2}{2R^3 \sec^3 \phi} R \sec^2 \phi d\phi dR = \frac{\mu_0 n I}{2} \int_{R_1}^{R_2} \int_{-\phi_0}^{\phi_0} \cos \phi d\phi dR$$

Evaluating the inside integral:

$$\int_{-\phi_0}^{\phi_0} \cos \phi d\phi = 2 \sin \phi_0 = \frac{2l}{\sqrt{l^2 + R^2}}$$

thus

$$B = \mu_0 n I \int_{R_1}^{R_2} \frac{l}{\sqrt{l^2 + R^2}} dR$$

substitute $R = l \tan \xi$, $dR = l \sec^2 \xi d\xi$, $\xi_1 = \arctan \frac{R_1}{l}$, $\xi_2 = \arctan \frac{R_2}{l}$:

$$B = \mu_0 n I \int_{\xi_1}^{\xi_2} \frac{l}{l \sec \xi} l \sec^2 \xi d\xi = \mu_0 n I l \int_{\xi_1}^{\xi_2} \sec \xi d\xi = \mu_0 n I l [\ln(\sec \xi + \tan \xi)]_{\xi_1}^{\xi_2}$$

which we evaluate as

$$B = \mu_0 n I l \ln \frac{\sec \xi_2 + \tan \xi_2}{\sec \xi_1 + \tan \xi_1} = \mu_0 n I l \ln \frac{\frac{\sqrt{R_2^2 + l^2}}{l} + \frac{R_2}{l}}{\frac{\sqrt{R_1^2 + l^2}}{l} + \frac{R_1}{l}}$$

$$= \mu_0 n I l \ln \frac{\frac{R_2}{R_1} + \sqrt{\left(\frac{R_2}{R_1}\right)^2 + \left(\frac{l}{R_1}\right)^2}}{1 + \sqrt{1 + \left(\frac{l}{R_1}\right)^2}}$$

with values $\alpha = R_2/R_1$, $\beta = l/R_1$, $\kappa = \mu_0$.

Note: evaluating the integral in the wrong order is very challenging!

(ii) Length of a single wire loop is $2\pi r$, thus length of the whole wire is

$$L = \int_{-l}^l \int_{R_1}^{R_2} (n \cdot 2\pi r) dr dx = n \int_{-l}^l \int_{R_1}^{R_2} 2\pi r dr dx = nV$$

which is

$$L = n(\pi R_2^2 - \pi R_1^2)(2l) = 2\pi n \left(\left(\frac{R_2}{R_1} \right)^2 - 1 \right) \left(\frac{l}{R_1} \right) R_1^3$$

with exponents $\mathbf{m_1 = 2, m_2 = 1, m_3 = 3}$.

(iii) Note that resistivity $\rho = \frac{1}{\sigma}$. Consider a coil of wire of length L and radius r ,

The dissipated power is: $P = I^2 R = I^2 \frac{L}{\sigma \pi r^2}$

$$\begin{aligned} B &= \mu_0 n l \beta R_1 \ln \frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{1 + \sqrt{1 + \beta^2}} = \mu_0 \beta \ln \frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{1 + \sqrt{1 + \beta^2}} (n^2 I^2 R_1^2)^{1/2} \\ &= \mu_0 \beta \ln \frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{1 + \sqrt{1 + \beta^2}} \left(\frac{n^2 P \sigma \pi r^2}{2\pi n (\alpha^2 - 1) \beta R_1^3 R_1^2} \right)^{1/2} \\ &= \mu_0 \beta \ln \frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{1 + \sqrt{1 + \beta^2}} \left(\frac{1}{2\pi (\alpha^2 - 1) \beta} \right)^{1/2} \left(\frac{P \lambda \sigma}{R_1} \right)^{1/2} \end{aligned}$$

which is the desired form.

9) (a) The angular frequency of oscillation is

$$\omega = \sqrt{\frac{k}{M_1 + M_2}}$$

where k is the spring constant.

The maximum acceleration of the top block is thus $a = A\omega^2$ where A is the amplitude. This corresponds to the maximum friction force $F = M_2 a = \mu_s M_2 g$ so the maximum amplitude is

$$A = \mu_s g / \omega^2$$

(b) Conservation of energy:

$$GPE = Mg \left(\frac{L}{2} \right) = KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{1}{3} M L^2 \right) \omega^2$$

thus

$$\omega = \sqrt{3g/L}$$

and the eraser hits the table at speed $v = L\omega = \sqrt{3gL}$.

10) (a) Let the observer move with speed $\beta_o = u/c$. The muon's relative speed is

$$\beta_r = \frac{\beta - \beta_o}{1 - \beta\beta_o}$$

We want to find when $\beta_r < 0$. The numerator is always positive (since $\beta > 1 > \beta_o$) while the denominator is negative when $\beta > \beta_o^{-1} \approx 1 - 1.48 \times 10^{-5}$ so the critical speed is

$$\mathbf{u = (1 - 1.48 \times 10^{-5})c}$$

(b) (i) Only the rod component parallel to direction of motion undergoes length contraction. In the observer's frame,

$$L_x = \sqrt{3} \text{ m}, L_y = 1.00 \text{ m}$$

So in the proper frame,

$$L'_x = \gamma L_x = \frac{\sqrt{3}}{\sqrt{1 - 0.995^2}} \text{ m} = 17.34 \text{ m}, L'_y = L_y = 1.00 \text{ m}$$

and it has proper length

$$L_0 = \sqrt{L'^2_x + L'^2_y} = \mathbf{17.37 \text{ m}}$$

(ii) $\theta_0 = \arctan L'_y/L'_x = \arctan 1/17.34 = \mathbf{0.0576 \text{ rad} = 3.3^\circ}$